Essays

|!|BCEEA0210000001|!| ★★ **Cross topic**

Green plants are autotrophs while humans are heterotrophs.
Compare how green plants and humans obtain carbohydrates, transport these molecules in their bodies and store them as energy reserve. (11 marks)

##

How green plants and humans obtain carbohydrates: (Max. 2)

- Green plants can capture light energy and synthesize glucose using carbon dioxide and water. (1)
- Green plants can use the glucose formed from photosynthesis to make other types of carbohydrates such as sucrose, starch and cellulose. (1)
- Humans can obtain carbohydrates from organic food. Complex carbohydrates (e.g. starch) in ingested food are digested into simple monosaccharides (e.g. glucose) for absorption. (1)

How green plants and humans transport carbohydrates in their bodies: (Max. 4)

- In plants, carbohydrates are transported in the form of sucrose. (1)
- Sucrose is transported from the leaves to other parts of the plants (e.g. roots, fruits and storage organs) through the phloem tissue. (1)
- Sucrose can also be transported from storage organs to the leaves and other growing regions. (1)
- The transport of sucrose through the phloem is multidirectional. (1)
- After carbohydrates are digested in the human body, monosaccharides are absorbed into the blood and transported by blood to various parts of the body. (1)
- Blood flow is maintained by the pumping action of the heart. (1)
- Blood flows in a branching network of blood vessels and circulates unidirectionally from the heart in a specific route and returns to the heart again. (1)

Storage of carbohydrates as energy reserve: (Max. 2)

- In plants, carbohydrates are often stored in the form of starch (1)
  in the stems, roots and underground storage organs. (1)
- In the human body, carbohydrates are often stored in the form of glycogen (1)
  in muscles and the liver. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

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|!|BCEEB0210000002|!| ★ **Cross topic**

Discuss how the structure of a dicotyledonous green leaf is related to certain vital functions of the plants. (12 marks)

##

The leaves of a plant are the main sites of gas exchange and photosynthesis. (1)

Adaptations of the leaves for gas exchange and photosynthesis: (Max. 8)

- Leaves are usually broad and flat. (1)
  This provides a large surface area for the exchange of gases and for absorbing light. (1)
- The leaf blade is supported by the midrib and veins so that it spreads out to maximize the surface area exposed to sunlight. (1)
- Leaves are thin. (1)
  This shortens the distance for light and gases to reach the mesophyll cells. (1)
- The cuticle and epidermis are transparent so that light can penetrate and reach the photosynthetic mesophyll beneath them. (1)
- The presence of stomata in the epidermis allows gases to pass into and out of the leaves. (1)
- The guard cells control the opening and closing of stomata, thus regulating the rate of gas exchange. (1)
  When light is not available for photosynthesis, the stomata close to reduce the loss of water vapour from the leaves. (1)
- The palisade mesophyll is located in the upper part of a leaf, facing the Sun directly. The cells are closely packed and contain numerous chloroplasts. (1)
  These help maximize the amount of light absorbed for photosynthesis. (1)
- The spongy mesophyll cells are loosely packed, with lots of air spaces among them. This allows gases to diffuse freely inside the leaves. (1)
- A film of water is present on the surface of spongy mesophyll cells. Gases can first dissolve in it and readily diffuse into the cells. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Applying concepts	9
Communication	3

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|!|BCEEA0210000003|!| ★★

With reference to the main stages of photosynthesis, explain what factors limit the rate of photosynthesis. (12 marks)

##

Photochemical reactions: (Max. 3)

- Light energy is absorbed by chlorophyll and its electrons are excited to a higher energy level. (1)
- As electrons pass through a series of electron carriers, energy is released. (1)
- The energy released is used to synthesize ATP through photophosphorylation. (1)
- The energy released is also used to split water molecules into oxygen and hydrogen. (1)
- Hydrogen ions combine with NADP, forming NADPH. (1)

The rate of photochemical reactions is limited by low light intensity. Less ATP and NADPH can be produced and enter the Calvin cycle. (1#)

The Calvin cycle: (Max. 3)

- Carbon dioxide combines with a 5-C carbon dioxide acceptor, forming two molecules of a 3-C compound. (1)
- The 3-C compound is reduced by NADPH, forming triose phosphate. (1)
- For every 12 molecules of triose phosphate formed, two of which is used to form a glucose molecule. (1)
- The remaining triose phosphate is used to regenerate the 5-C carbon dioxide acceptor so that the cycle can repeat. (1)

The rate of the Calvin cycle is limited by low light intensity, because it requires ATP and NADPH from the photochemical reactions. (1#)

The rate of the Calvin cycle is also limited by low concentration of carbon dioxide. Fewer 3-C compounds can be formed from carbon dioxide fixtion and enter the Calvin cycle. (1)

Temperature is a limiting factor of photosynthesis because many reactions are catalysed by enzymes. (1)

Communication (Max. 3)

Marks should be awarded once only.

(12 marks)

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Skills assessed	Marks
Applying concepts	9
Communication	3